

Targeting SLC7A11-mediated cysteine metabolism for the treatment of trastuzumab resistant HER2 positive breast cancer

Reviewed Preprint

v2 • April 23, 2025

Revised by authors

Reviewed Preprint

v1 • February 5, 2025

Yijia Hua, Ningjun Duan , Chunxiao Sun, Fan Yang, Min Tian, Yanting Sun, Shuhan Zhao, Jue Gong, Qian Liu, Xiang Huang, Yan Liang, Ziyi Fu, Wei Li, Yongmei Yin 

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China • Gusu School, Suzhou Municipal Hospital, The Affiliated Suzhou Hospital of Nanjing Medical University, Suzhou, China • Department of General Surgery, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China • Jiangsu Key Lab of Cancer Biomarkers, Prevention and Treatment, Collaborative Innovation Center for Personalized Cancer Medicine, Nanjing Medical University, Nanjing, China

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This study provides **compelling** evidence that SLC7A11 may serve as a potential therapeutic target for trastuzumab-resistant HER2-positive breast cancer. While the findings are well-supported by robust data, the study could have been further strengthened by incorporating additional cell line experiments and providing more detailed clarification on patient sample selection. Nevertheless, this **valuable** work represents a significant contribution and will be of considerable interest to researchers in the field of breast cancer.

<https://doi.org/10.7554/eLife.103953.2.sa3>

Abstract

Trastuzumab resistance remains a challenge for HER2 positive breast cancer treatment. Targeting metabolic reprogramming would provide novel insights for therapeutic strategies. Here, we integrated metabolomics, transcriptomics and epigenomics data of trastuzumab sensitive and primary resistant HER2 positive breast cancer to identify metabolic alterations. Aberrant cysteine metabolism was discovered in trastuzumab primary resistant breast cancer at both circulating and intracellular levels. The inhibition of SLC7A11 and cysteine starvation could synergize with trastuzumab to induce ferroptosis. Mechanistically, increased H3K4me3 and decreased DNA methylation enhanced SLC7A11 transcription and cystine uptake in trastuzumab resistant breast cancer. The regulation of epigenetic modifications modulated cysteine metabolism and ferroptosis sensitivity. These results revealed an innovative approach for overcoming trastuzumab resistance by targeting specific amino acid metabolism.

Introduction

Accounting for 15%-20% of all breast cancer cases, human epidermal growth factor receptor 2 (HER2) positive breast cancer has posed great threats to the health of women worldwide(1, 2). The overexpression of HER2 results in aberrant breast cancer cell proliferation, invasion, and progression(3). As a monoclonal antibody targeting HER2, trastuzumab has been recognized as the foundational medication in HER2 positive breast cancer treatment and provided remarkable clinical benefits for numerous patients(4). Around 50% HER2 positive breast cancer patients would experience trastuzumab resistance and disease progression during or after treatment settings(5). Current theories maintained that trastuzumab resistance mainly resulted from HER2 mutation(6), antigenic epitope variation(7), compensatory pathway activation(8) and tumor heterogeneity(9). However, there still lacks a comprehensive theory to thoroughly explain the mechanisms behind trastuzumab resistance, and exploring effective targets to overcome resistance remains a major topic in the field of breast cancer research(3).

Metabolic reprogramming has been acknowledged as one crucial cancer hallmark(10). Traced back to the first study carried out by Otto Warburg, decades of studies have indicated that glycolysis, tricarboxylic acid (TCA) cycle, lipid metabolism, amino acid metabolism, as well as other processes collectively modulate the development and function of cancer cells and tumor microenvironment (TME) components(11–13). Several studies have revealed that different breast cancer subtypes featured diverse metabolic patterns(14, 15). Targeting specific enzymes or modulating nutrients uptake might normalize the TME and enhance breast cancer treatment efficacy(16, 17). Yet previous studies often focused on the overall metabolic subtypes of breast cancer, with little attention given to abnormal metabolic changes and underlying genomic alterations induced by specific drug resistance. This might lead to a poor understanding on the tumor heterogeneity in drug resistant breast cancer cells.

Epigenetic modifications play crucial roles in all stages of cancer development and contribute to cancer cell survival selection(18). Alterations in chromatin structure, DNA methylation and histone modifications provide tumor growth advantage and increase phenotypic variability(19). During the formation of drug resistance, epigenetic dysregulation modulates key genes transcription associated with treatment tolerance, leading to resistant clonal derivatives and disease progression(20). Previous studies suggested that COMPASS complex inactivation suppressed tumor suppressor genes transcription and accelerated breast cancer tumorigenesis(21). Aberrant DNA methylation resulted in abnormal oncogenic signaling activation and drug resistance acquisition(22). Targeting epigenetic reprogramming might provide novel treatment options for overcoming breast cancer drug resistance.

By integrating metabolomic, transcriptomic and epigenomic data, we identified metabolic alternations, especially cysteine-associated pathways, accompanied with the epigenetic changes during the development of primary trastuzumab resistance of HER2 positive breast cancer (**Figure 1A**). Based on these discoveries, targeting certain metabolism-related protein, epigenetic modification or nutrition could all be possible ways to overcome trastuzumab resistance. In all, our study provided novel insights and potential solutions on primary trastuzumab resistance of HER2 positive breast cancer.

Methods

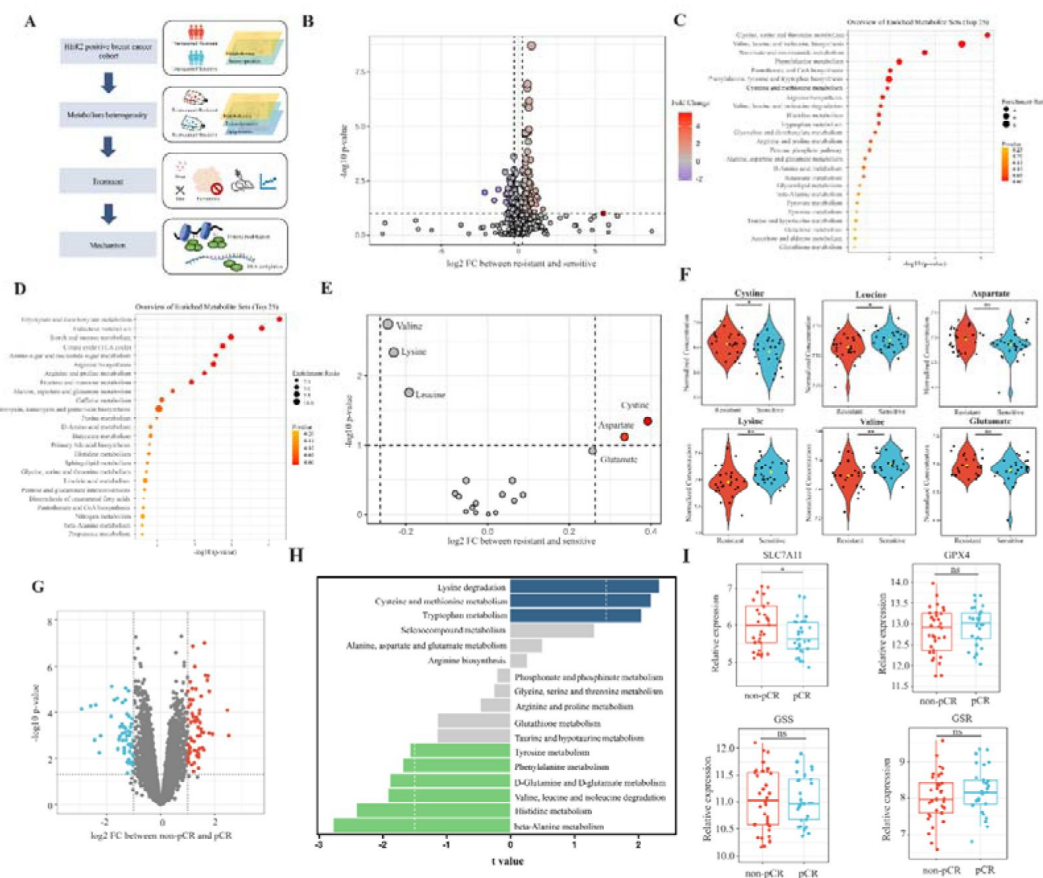


Figure 1.

Analyses of amino acids metabolic patterns in trastuzumab primary resistant and sensitive HER2 positive breast cancer patients.

(A) Workflow of analyses performed in this study. (B) Volcano plot of different circulating metabolites in the comparison between trastuzumab primary resistant and sensitive patients. (C) Enrichment analysis of circulating metabolites downregulated in trastuzumab primary resistant patients. (D) Enrichment analysis of circulating metabolites upregulated in trastuzumab primary resistant patients. (E-F) Volcano plot (E) and violin plots (F) of different circulating protein-construction amino acids in the comparison between trastuzumab primary resistant and sensitive patients. (G) Volcano plot of different genes between non-pCR and pCR patients in trastuzumab-based neoadjuvant treatment (I-SPY2, GSE181574). (H) Amino acids metabolic pathway analysis of non-pCR and pCR patients. (I) Relations between cysteine metabolic genes and trastuzumab treatment outcomes. Significances were determined by two-tailed unpaired t-test (F, I). ns, $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

Patient cohort and blood samples collection

One breast cancer patient cohort from Jiangsu Province Hospital was embedded in this study, which consists of 26 primary trastuzumab resistant patients defined as disease recurrence during or after (≤ 12 months) trastuzumab adjuvant treatment and 26 trastuzumab sensitive patients that benefited from trastuzumab adjuvant treatment more than 12 months(23 [23](#)).

All blood samples were collected before the first trastuzumab treatment cycle. Samples collection was under the approval of the Ethics Committee and Institutional Review Board of Jiangsu Province Hospital. Each patient provided written informed consent for sample and data use.

Cell culture

Human breast cancer cell lines JIMT1, SKBR3 and human breast epithelial cell line MCF10A were obtained from American Type Culture Collection (ATCC). JIMT1 and SKBR3 cell lines were cultured in high-glucose Dulbecco's modified eagle medium (DMEM; Gibco, 11965092) with 10% fetal bovine serum (FBS; Gibco, 16140071) and 1% penicillin streptomycin (Gibco, 15140122) at 37LJ with 5% CO₂. MCF10A was cultured in mammary epithelial cell growth medium (MEGM) supplemented with bovine pituitary extract, human epidermal growth factor, hydrocortisone, gentamicin sulfate-amphotericin, and insulin (Lonza, CC-3150) with 100 ng/ml cholera toxin (Sigma, C8052) at 37LJ with 5% CO₂. Cell culture with cysteine starvation was performed by cystine/cysteine-deficient DMEM (BIOTREE) and 1% penicillin streptomycin at 37°C with 5% CO₂.

DNA constructs, plasmids, and transfection

Short interfering RNA (siRNA) targeting ASH2L, SLC7A11 and GPX4 were presented in Supplementary Table S1. The design of small guide RNA (sgRNA) was based on the relevant methylated region and conducted via the online tool CRISPick(24 [24](#), 25 [25](#)). Sequences of sgRNA were indicated in Supplementary Table S2. dCas9-DNMT3A was obtained from Addgene (#100090) (26 [26](#)).

Transfection was conducted by using Lipofectamine 3000 (Invitrogen, L3000015) according to the manufacturer's protocol. In brief, cells were cultured in a six-well plate at 70%-90% confluency. For RNA interference, 2500 ng siRNA and 3.75 μ l Lipofectamine 3000 reagents were mixed and then added in each well. For CRISPR-based epigenetic editing, dCas9-DNMT3A plasmid and sgRNA were first mixed in 1:2 ratio, then mixed with 3.75 μ l Lipofectamine 3000 reagent and 5 μ l P3000 reagent in each well.

Protein extraction and Western blots

Cells were collected and then lysed in SDS lysis buffer with phenylmethyl-sulfonyl fluoride (PMSF; Servicebio, G2008) and phosphatase Inhibitor cocktail (Servicebio, G2007) on ice. Proteins were separated in SDS-PAGE and transferred to PVDF membranes (Millipore, IPVH00010). After blocked, membranes were incubated with specific primary and secondary antibodies, and then visualized with imaging system and ECL chemiluminescence kit (Vazyme, E422-01). Primary antibodies included SLC7A11 (CST, 12691S), GPX4 (CST, 52455S), glutathione synthetase (Abcam, ab124811), glutathione reductase (Abcam, ab124995), ASH2L (CST, 5019T), H3K4me3 (Abcam, ab213224), GAPDH (proteintech, HRP-60004), and H3 (Abcam, ab1791). The secondary antibody was horseradish peroxidase (HRP) conjugated goat anti-rabbit IgG(H+L) (ImmunoWay, RS0002).

GSH/GSSG ratio assay

GSH/GSSG ratio was measured with a GSH and GSSG assay kit (Beyotime, S0053). Cells were lysed with protein removing reagent solution and several freeze-thaw cycles. After centrifuged, supernatants were collected and incubated with DTNB and NADPH.

The absorbance at 412 nm was detected to measure the GSH/GSSG ratio with standard curve.

C11 BODIPY (581/591) assay

Cells were collected and incubated with 5 μ M C11 BODIPY (581/591) (Invitrogen, D3861) for 30 minutes at 37 °C. The oxidized and reduced state were detected by flow cytometry, with excitation and emission wavelength at 488/510 and 581/591 nm, respectively.

DCFH-DA assay

Cells were collected and incubated with 10 μ M DCFH-DA (MCE, HY-D0940) for 30 minutes at 37 °C. The oxidized state was detected with laser excitation wavelength at 488 nm.

Cystine uptake assay

The cystine uptake ability was measured by using cystine uptake assay kit (DOJINDO, UP05). Cells were incubated with cystine analog solution and fluorescent probe. The fluorescence intensity was measured with excitation and emission wavelength at 490 and 535 nm.

Quantification of intracellular cysteine

The concentration of intracellular cysteine was detected by using a cysteine quantification kit (Solarbio, BC0185). Cells were lysed and incubated with solution containing phosphotungstic acid. The absorbance at 600 nm was detected to measure intracellular cysteine levels.

ChIP

Chromatin extraction and the following immune precipitation were performed with chromatin extraction kit (Abcam, ab117152) and ChIP kit (Abcam, ab117138) according to manufacturer's protocols. Cells were harvested and treated with 1% formaldehyde first, then sonication was applied on cross-linked chromatin to generate 200-1000 bp fragments. Chromatin fractions enriched with certain histone modifications were captured by antibodies (H3K27me3 (CST, C36B11), H3K4me3 (Abcam, ab213224) and IgG) and released by protease digestion.

RT-qPCR was applied to quantify the enrichment of specific type of histone modification at certain chromatin regions with ChamQ SYBR qPCR Master Mix (Vazyme, Q311). While VAHTS® Universal DNA Library Prep Kit for Illumina (Vazyme, ND607) and Illumina Novaseq 6000 platform were applied for sequencing library preparation and the following sequencing. All primers used for ChIP-qPCR were presented in Supplementary Table S3.

MeDIP

Methylated DNA immune precipitation was performed with MeDIP kit (Abcam, ab117135) according to manufacturer's protocols followed by chromatin extraction as ChIP protocol. After cross-linking and sonication, methylated chromatin fractions were enriched by anti-5-Methylcytosine (5-mC) antibody (CST, 28692). RT-qPCR was applied to quantify the enrichment of 5-mC at certain chromatin regions. All primers used for MeDIP-qPCR were presented in Supplementary Table S3.

Dot Blots

Total genomic DNA was extracted by using nucleic acid isolation kit (Vazyme, DC112). Denatured DNA was spotted on a nitrocellulose membrane, and blocked with 5% skim milk in TBST (20 mM Tris, 150 mM NaCl, and 0.1% Tween-20). Anti-5-mC primary antibody and HRP-conjugated secondary antibody together with ECL chemiluminescence kit were applied for total methylated cytosine visualization.

Mice xenograft model construction

Six-week-old female BALB/c nude mice were obtained from Beijing Vital River Laboratory Animal Technology, housed in a 12-hour light-dark lighting condition with 20–22 °C temperature and 60 ± 10% humidity. The animal diet and drinking water followed the national standard for experimental animal feed. The diet underwent Cobalt-60 irradiation sterilization and the drinking water followed purification and sterilization.

All animal experiments were conducted according to the review and approval of Institutional Animal Care and Use Committee in Nanjing Medical University (IACUC-2204057). 3×10^6 JIMT1 cells were injected subcutaneously into the mammary fat pad region to generate xenograft models.

The mice were divided into 6 groups: (1) DMSO treatment; (2) trastuzumab (Roche) treatment (10mg/kg, injected intraperitoneally twice a week); (3) erastin (HY-15763) treatment (40mg/kg, injected intraperitoneally every other day); (4) cysteine starvation; (5) trastuzumab treatment plus cysteine starvation; (6) trastuzumab plus erastin treatment. Cysteine starvation was performed by utilizing cystine/cysteine deficient diet (Xietong Bio). Treatment was initiated when tumor volume reached 50mm³. Tumor sizes and mice weight were measured three times a week. Tumor volumes were calculated as $0.5 \times L \times W^2$ (L: the longest dimension, W: the perpendicular dimension). The tumor growth was compared by paired individual fold change of tumor volumes.

Tumor and spleen cell composition analysis

Tumor samples were digested in serum-free RPMI with 10mg/ml collagenase I (Sigma, C0130) and 1mg/mg DNase I (Roche, 10104159001) for 30–60 minutes at 37 °C. Spleen samples were gently crushed. Both of them were filtered in 70µm strainers, followed by red blood cell removal with RBC lysis buffer (eBioscience, 00-4333-57), and then resuspended in flow cytometry staining buffer. Fc receptor was blocked with mouse FcR blocking reagent (Miltenyi, 130-092-575).

Fluorescent antibodies including CD45 (clone 30-F11, eBioscience, 48-0451-82), CD3 (clone 145-2C11, BioLegend, 100353), CD49b (clone DX5, BioLegend, 108907), CD11b (clone M1/70, eBioscience, 56-0112-82), CD27 (clone LG.3A10, BioLegend, 124216), CD335 (NKp46) (clone 29A1.4, eBioscience, 17-3351-82) and granzyme B (clone NGZB, eBioscience, 11-8898-82) together with fixable viability dye (eBioscience, 65-0865-14) were chosen for cell staining. CytoFLEX (Beckman Coulter) and FlowJo software were applied for fluorescence measurement and analysis.

Immunohistochemical staining

After formalin fixation and paraffin embedding, tissue sections were cut and mounted on slides followed by deparaffinizing, rehydrating and antigen retrieval. Samples were then stained with antibodies including SLC7A11 (Abcam, ab307601), GPX4 (Abcam, ab125066), 4-Hydroxynonenal (4-HNE) (Abcam, ab48506), Malondialdehyde (MDA) (Abcam, ab243066), and caspase-3 (Abcam, ab32150).

Differential transcriptome analyses

The human hg38 genome was utilized for RNA sequencing reads alignment. Differential expression analyses were measured by the DESeq2 package in R. Significant differential expression was defined with a threshold of $P \text{ adj} \leq 0.05$ and $|\log_2 \text{ fold change}| \geq 0.5$.

Differential metabolome analyses

Metabolomics data was analyzed by the MetaboAnalystR package in R(27 [↗](#)). Pathway enrichment analyses was performed according to KEGG databases. Significant difference was defined with a threshold of $P_{\text{adj}} \leq 0.05$ and $|\log_2 \text{fold change}| \geq 0.5$.

Differential epigenome analyses

Sequencing reads of ChIP and whole-genome bisulfite sequencing (WGBS) were aligned to the human hg38 genome. Differential methylation analyses was performed by deepTools (version 3.5.5)(28 [↗](#)). Differentially methylated regions (DMR) were identified by Bismark (version 0.24.2) (29 [↗](#)). Significant difference was defined with a threshold of $P_{\text{adj}} \leq 0.05$ and $|\log_2 \text{fold change}| \geq 0.5$.

Genomics analyses

The human hg38 genome was utilized to align WGS sequencing reads. Germline single nucleotide polymorphisms (SNP) and Indels were identified by GATK (v 4.5.0.0). The annotation of germline mutations was obtained by utilizing ANNOVAR.

Statistical analyses

All statistical analyses were performed by utilizing GraphPad Prism (version 10.1.2) and R (version 4.3.3). All data were represented as the mean $\pm$ standard error (SE) or the mean $\pm$ SEM from a minimum of three independent experiments. Two-tailed Student's t-test was applied to determine statistical in comparisons between two groups and ANOVA test in multiple groups. The threshold for statistical significance is $P < 0.05$, and was marked as * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Results

Different amino acids metabolic patterns between trastuzumab sensitive and primary resistant HER2 positive breast cancer patients

To investigate metabolic patterns of HER2 positive breast cancer patients with different trastuzumab response, we obtained and analyzed the plasma metabolites data of both trastuzumab sensitive and primary resistant patients (26/26) with good interclass correlations (Figure S1A-C [↗](#)). Of them, 203 metabolites (130 upregulated and 73 downregulated in primary resistant groups) were identified with different abundances among two patient groups (Figure 1B [↗](#) and S1D [↗](#)). KEGG-based pathway analyses suggested the major enrichment in amino acids metabolism. For instance, glycine, serine and threonine metabolism, valine, leucine and isoleucine biosynthesis, as well as cysteine and methionine metabolism were reduced in primary resistant patients; while arginine, proline, alanine, aspartate and glutamate metabolism were enhanced (Figure 1C [↗](#) and 1D [↗](#)).

By comparing the abundance of 20 common protein-construction amino acids, we realized that cystine and aspartate were higher while valine, lysine and leucine were lower in primary resistant groups. However, glutamate kept stable in both groups (Figure 1E [↗](#) and 1F [↗](#)). Non-protein amino acids also displayed difference among these two groups. 2-aminoisobutyric acid and 4-aminobutyric acid (GABA) were lower in resistant groups, while citrulline, kynurenine, ornithine and taurine levels were similar (Figure S1E [↗](#)).

We then analyzed transcriptomic data of pre-treatment breast cancer tumors in I-SPY2 trial⁽³⁰⁾ to further explore metabolic gene alterations, in which HER2 positive breast cancer patients received trastuzumab plus paclitaxel, trastuzumab combined with pertuzumab and paclitaxel or T-DM1 plus paclitaxel during neoadjuvant settings. By comparing patients with different responses to neoadjuvant trastuzumab-based regimens (pathological complete response [pCR] and non-pCR), we found 1542 genes upregulated and 1350 genes downregulated in non-pCR groups (Figure S1F and 1G). Based on KEGG datasets, amino acids metabolism was found dysregulated in patients with poor responses to trastuzumab, including aberrant lysine degradation, cysteine and methionine metabolism, beta-alanine metabolism and histidine metabolism (Figure 1H).

As cystine was one of the most significant amino acids in both metabolomic and transcriptomic analyses, we investigated relations between cystine/cysteine metabolic key genes and trastuzumab treatment outcomes. Patients with non-pCR featured higher expression of SLC7A11, while no differences were found in GPX4, GSS and GSR expression (Figure 1I), indicating SLC7A11 might become a potential target for trastuzumab primary resistant patients.

In general, trastuzumab primary resistant HER2 positive breast cancer patients featured distinct amino acids metabolic patterns, especially increased plasma cystine levels and higher expression of SLC7A11.

Distinct cysteine metabolism in trastuzumab primary resistant HER2 positive breast cancer

To comprehensively compare the metabolic heterogeneity between trastuzumab sensitive and primary resistant HER2 positive breast cancer, we obtained and analyzed both metabolomic and transcriptomic data of trastuzumab sensitive SKBR3 and trastuzumab resistant JIMT1 cell lines.

Significant differences were observed in metabolic data with good quality (Figure S2A and S2B). Specifically, 181 upregulated and 235 downregulated hydrophilic metabolites accompany 546 upregulated and 412 downregulated lipophilic metabolites were found in JIMT1 cells compared to SKBR3 cells (Figure 2A and S2C). For hydrophilic metabolites, enrichment analyses revealed that glycine, serine and threonine metabolism were upregulated in JIMT1, while cysteine and methionine metabolism were downregulated (Figure 2B and 2C). Enrichment analyses based on lipophilic metabolites showed that linoleic acid metabolism, fatty transportation and glycerophospholipid catabolism were upregulated in JIMT1, but glycerolipid metabolism, phospholipid biosynthesis and retinoid metabolism downregulated (Figure S3A and S3B).

Differential gene expression could also be uncovered from transcriptomic data. 4235 downregulated genes and 4923 upregulated genes were identified in the comparison between JIMT1 and SKBR3 cells (Figure S4A). Based on KEGG datasets, activities of cysteine and glutathione metabolism pathways were found dysregulated in JIMT1 cells (Figure 2D). More importantly, the joint pathway analyses combining metabolomic and transcriptomic data indicated that JIMT1 featured aberrant glutathione metabolism (Figure 2E).

The extracellular cystine is mainly imported by SLC7A11 for subsequent protein synthesis, glutathione (GSH) production and oxidative catabolism (Figure 2F). Both transcriptomic and protein evidence revealed the upregulation of SLC7A11 and glutathione synthetase (GSS) as well as the downregulation of GPX4 and glutathione reductase (GSR) in JIMT1 cells (Figure S4B and 2G), together with the characteristics that JIMT1 cells showed higher cysteine uptake ability and cytoplasmic cysteine level but lower glutathione/glutathione disulfide (GSH/GSSG) ratio than SKBR3 cells (Figure 2H-J), indicating the probably increased transsulfuration activity and decreased γ -glutamyl-peptides synthesis activity in JIMT1 cells (Figure S4C). These results suggested that HER2 positive breast cancer with primary trastuzumab resistance might prefer to

consume more cystine and feature intracellular cysteine metabolic reprogramming, showing high consistency with metabolic characteristics of relevant breast cancer patients and may associate with trastuzumab treatment response.

Stimulate ferroptosis sensitivity of trastuzumab primary resistant HER2 positive breast cancer

Since cysteine metabolism plays key role in maintaining cellular oxidative-reductive balance, disruption of which may turn breast cancer cells into a more sensitive state to oxidative damage, including ferroptosis^(31, 32). We first downregulated the expression of SLC7A11 and GPX4 in JIMT1 by siRNAs, and the lipid peroxidation of ferroptosis was found increased by measuring BODIPY-C11 (**Figure 3A** and **3B**). Similar results were also detected in SKBR3(**Figure S5A and S5B**). We then detected the sensitivity of JIMT1 and SKBR3 to SLC7A11 inhibitor (erastin) and GPX4 inhibitor (RSL3) and found JIMT1 was more sensitive to these two inhibitors (**Figure S5C and S5D**). The deprivation of cysteine could suppress the growth of both JIMT1 and SKBR3, but JIMT1 showed more tolerance to this nutrition starvation (**Figure S5E**).

12 hours treatment of SLC7A11 inhibitor (erastin), GPX4 inhibitor (RSL3) and even cysteine starvation were all able to induce ferroptosis in both JIMT1 and SKBR3 cells (**Figure 3C** and **3D**). However, JIMT1 cells indicated more rapidly accumulation of lipid peroxidation under the treatment of erastin, which could be detected by both increased BODIPY-C11 and DCFH-DA (**Figure 3E** and **3F**). Compared with SKBR3, JIMT1 cells were also more sensitive to RSL3 but more durable to cysteine starvation (**Figure 3G** and **3H**), which might be explained by its stronger transsulfuration activities and decreased γ -glutamyl-peptides synthesis (**Figure S5F**). The utilization of ferroptosis inhibitor Fer-1 could suppress the lipid peroxidation resulted from erastin, RSL3 and cysteine starvation in both JIMT1 and SKBR3, which also indicated the role of cysteine metabolism in modulating HER2 positive breast cancer ferroptosis (**Figure S5G and S5H**).

To understand whether trastuzumab would modulate the ferroptosis sensitivity, we treated JIMT1 cells with trastuzumab for different durations. We found that with a prolonged exposure of trastuzumab for 72 hours, the total amount of SLC7A11 in JIMT1 was decreased (**Figure 3I**). Correspondingly, cystine uptake ability was suppressed after trastuzumab treatment for 48 and 72 hours (**Figure 3J**). In addition, cells underwent trastuzumab treatment combining with erastin or cysteine starvation could demonstrate stronger lipid peroxidation and enhanced ferroptosis (**Figure 3K**).

Restraining cysteine metabolism inhibited tumor growth in vivo

To further investigate the effects of SLC7A11 inhibition or cysteine limitation on trastuzumab primary resistant HER2 positive breast cancer, JIMT1 xenograft mouse models were generated and then treated with different strategies including DMSO, trastuzumab, erastin, cysteine starvation, trastuzumab plus erastin, and trastuzumab plus cysteine starvation (**Figure 4A**).

Of them, trastuzumab plus erastin and trastuzumab plus cysteine starvation could effectively inhibit tumor growth, while cysteine starvation and erastin monotherapy only showed limited effects on tumor progression, demonstrating the better treatment responses of combination strategy (**Figure 4B, 4C** and **S6A**). In addition, almost no weight differences were observed among all groups during treatment, partially indicating the safety of such cysteine limitation treatment (**Figure 4D**).

The immune cell compositions in tumor and spleen of each group were measured to analyze the immune response induced by different treatment strategies (**Figure S6B and S6C**). The infiltration of NK cells (CD3-CD45+CD49b+) in tumors was increased only upon the combination of cysteine deprivation and trastuzumab (**Figure 4E**). Nearly no differences were found in tumor

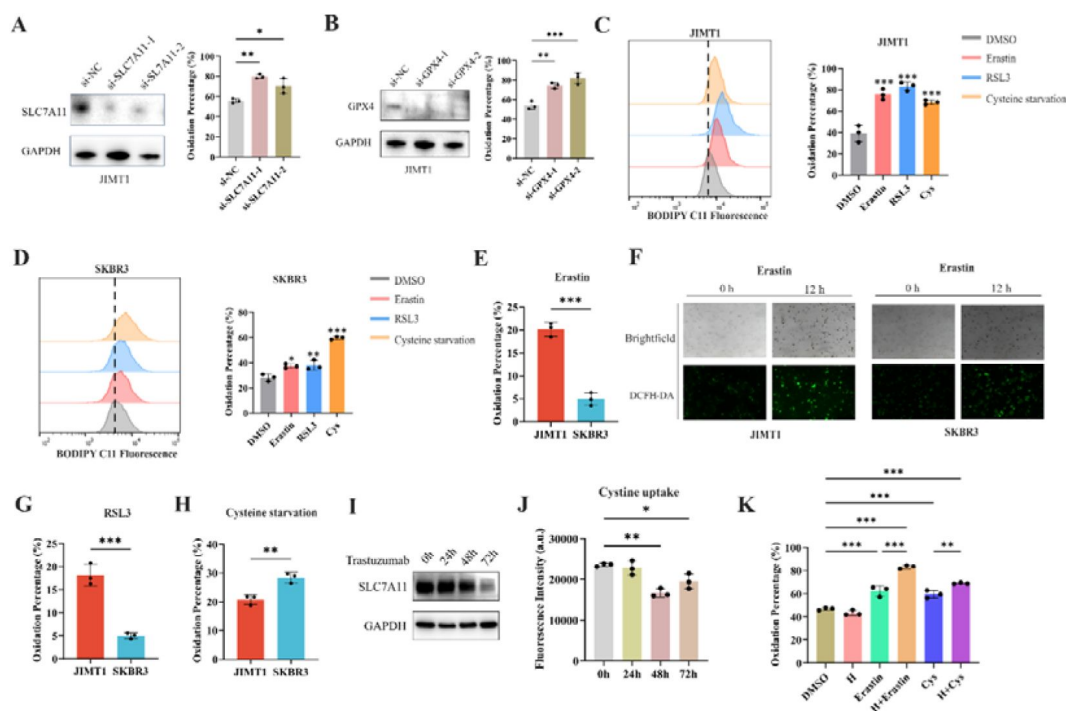


Figure 3.

Ferroptosis sensitivity of HER2 positive breast cancer with different trastuzumab response.

(A-B) Knockdown of SLC7A11 (A) and GPX4 (B) induced increased lipid peroxidation (BODIPY-C11) in JIMT1. (C-D) Treatment with erastin (10 μ m), RSL3 (1 μ m) and cysteine starvation for 12 hours induced lipid peroxidation (BODIPY-C11) in JIMT1 (C) and SKBR3 (D). (E-F) JIMT1 and SKBR3 featured different lipid peroxidation sensitivity to erastin by measuring BODIPY-C11 (E) and DCFH-DA (F). (G-H) JIMT1 and SKBR3 featured different lipid peroxidation sensitivity to RSL3 (G) and cysteine starvation (H). (I) SLC7A11 expression under different trastuzumab treatment time. (J) Cystine uptake ability under different trastuzumab treatment time. (K) The combination of erastin or cysteine starvation and trastuzumab increased lipid peroxidation in JIMT1. H, trastuzumab; Cys, cysteine starvation. Significances were determined by one-way ANOVA (A-D, J, K) and two-tailed unpaired t-test (E, G, H). ns, $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

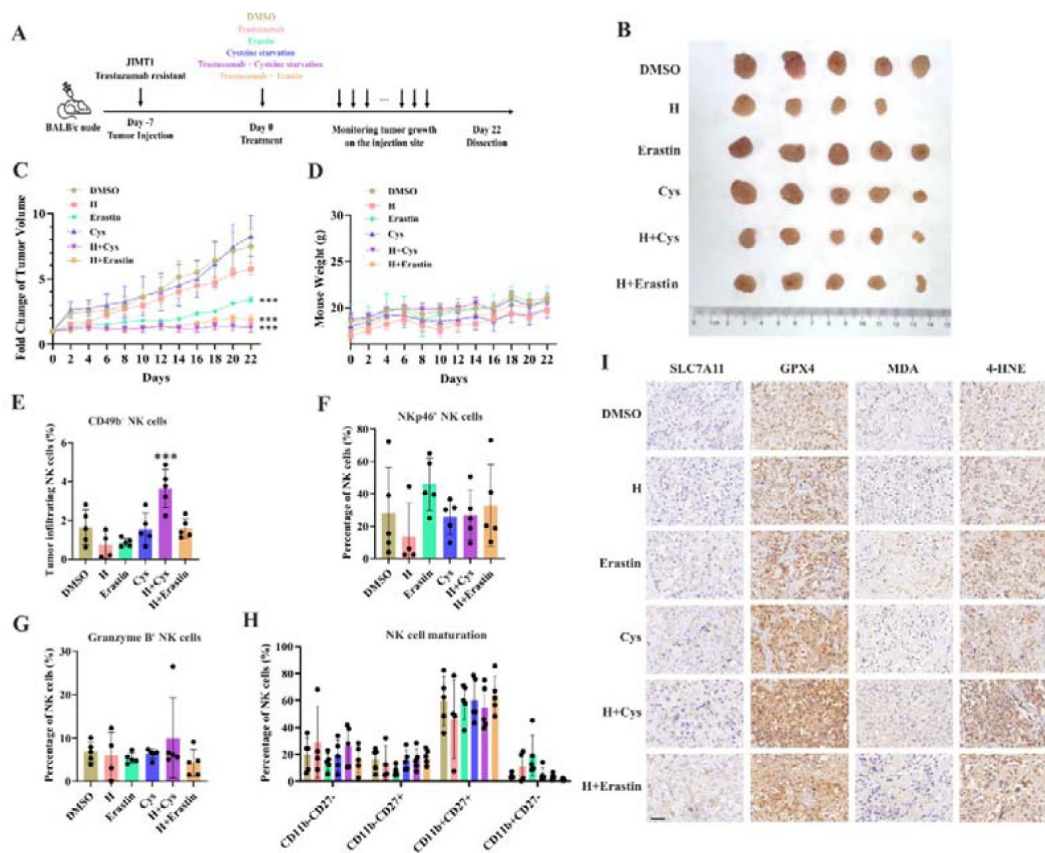


Figure 4.

Targeting cysteine metabolism synergizes with trastuzumab to induce ferroptosis.

(A) Schematic outline of treatment on cysteine metabolism in trastuzumab resistant breast cancer. JIMT1 cells were injected into the mammary fat pads of BALB/c nude mice at day -7. Treatment with DMSO, erastin, cysteine starvation, trastuzumab combined with or without erastin and cysteine starvation started at day 0 for 21 days. Tumor volume and mice weight were measured. The tumor growth was compared by paired individual fold change of tumor volumes. (B) Tumor of different treatment groups. (C) Fold change of tumor volume in different treatment groups. (D) Mice weight of different treatment groups. (E-H) Abundance of CD49b+ (E), Nkp46+ (F), Granzyme B+ (G) and different development status (H) tumor infiltrating NK cells. (I) Representing immunohistochemistry images of SLC7A11, GPX4, MDA and 4-HNE in different treatment groups. Scale bar, 50 μ m. H, trastuzumab; Cys, cysteine starvation. Significances were determined by two-way ANOVA (C, D) and one-way ANOVA (E-H). ns, $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

infiltrating NK cell activation (NKp46), toxic function (Granzyme B) and maturation status (CD11b, CD27) among all these groups (**Figure 4F-H**). While similar results were also observed in spleen NK cells (**Figure S6D-G**).

These results suggested a better effect of cysteine starvation than erastin in NK cell recruitment and activation, resulting a more powerful synergy with trastuzumab to increase the abundance of tumor infiltrating NK cells and show better inhibition of trastuzumab primary resistant breast cancer. As the anti-tumor effects of trastuzumab mainly relies on antibody dependent cellular cytotoxicity (ADCC) effects mediated by NK cells(33), this combination therapy might provide more effective approaches for HER2 positive breast cancer treatment.

We then analyzed the biomarkers of cysteine metabolism and cell death pathways in tumor samples of different treatment groups (**Figure 4I**). Erastin and cysteine deprivation monotherapy, or combined with trastuzumab, could upregulate SLC7A11 and GPX4 expression, as well as the abundance of MDA and 4-HNE. Ferroptosis might increase endoplasmic reticulum stress and activate p53 upregulated modulator of apoptosis (PUMA)(34). The combination of trastuzumab and erastin or trastuzumab and cysteine starvation might also modify apoptosis sensitivity. Apoptosis biomarker caspase-3 was found increasing in these synergic treatment groups (**Figure S6H**). These results suggested that targeting cysteine metabolism could induce ferroptosis in trastuzumab primary resistant HER2 positive breast cancer and might have a synergic effect with trastuzumab to inhibit breast cancer progression.

Aberrant promoter H3K4me3 regulates SLC7A11 gene expression and cysteine metabolism

Abnormal changes of histone modification play critical roles in tumor progression including drug resistance(35). Of them, altered methylations on promoter histone H3K4 and H3K27 show most important effects on gene expression(36).

By comparing the ChIP-seq data, we detected 3934 and 3925 H3K4me3 peaks were enriched in JIMT1 and SKBR3 respectively. These altered peaks primarily located among the promoter regions of genes (**Figure 5A** and **5A**), which have close relationships with biological processes such as nucleic acid transcription, protein modification, mRNA processing and ribosome processing (**Figure S8A**).

We also confirmed that 7048 and 7001 H3K27me3 peaks were enriched in JIMT1 and SKBR3 respectively, mainly enriching at not only promoter regions but also distal intergenic regions, and exons (**Figure 5B** and **5B**) of genes that involve in cell-cell adhesions, axon guidance, postsynaptic transmission, and neuron differentiation (**Figure S8B**).

Particularly, we uncovered the enrichment of H3K4me3 at the promoter regions of key cysteine transporter gene SLC7A11 in JIMT1 cells compared with SKBR3 cells while H3K27me3 signals were stable (**Figure 5C**), showing the consistency with its high expression in JIMT1 cells.

By inhibiting the expression of ASH2L, one crucial subunit of COMPASS complex(37), we downregulated not only the overall H3K4me3 level but also the signal at SLC7A11 promoter in JIMT1 cells, and successfully decreased the expression of SLC7A11 (**Figure 5D-F**). Meanwhile, diminished downstream cysteine uptake led to increased lipid peroxidation, the key symbol of ferroptosis (**Figure 5G** and **5H**).

These results suggested that H3K4me3 may play a more important role in modulating SLC7A11 transcription and cysteine metabolism in JIMT1 than H3K27me3. Targeting H3K4me3 could regulate SLC7A11 expression, enhancing lipid peroxidation and inducing ferroptosis.

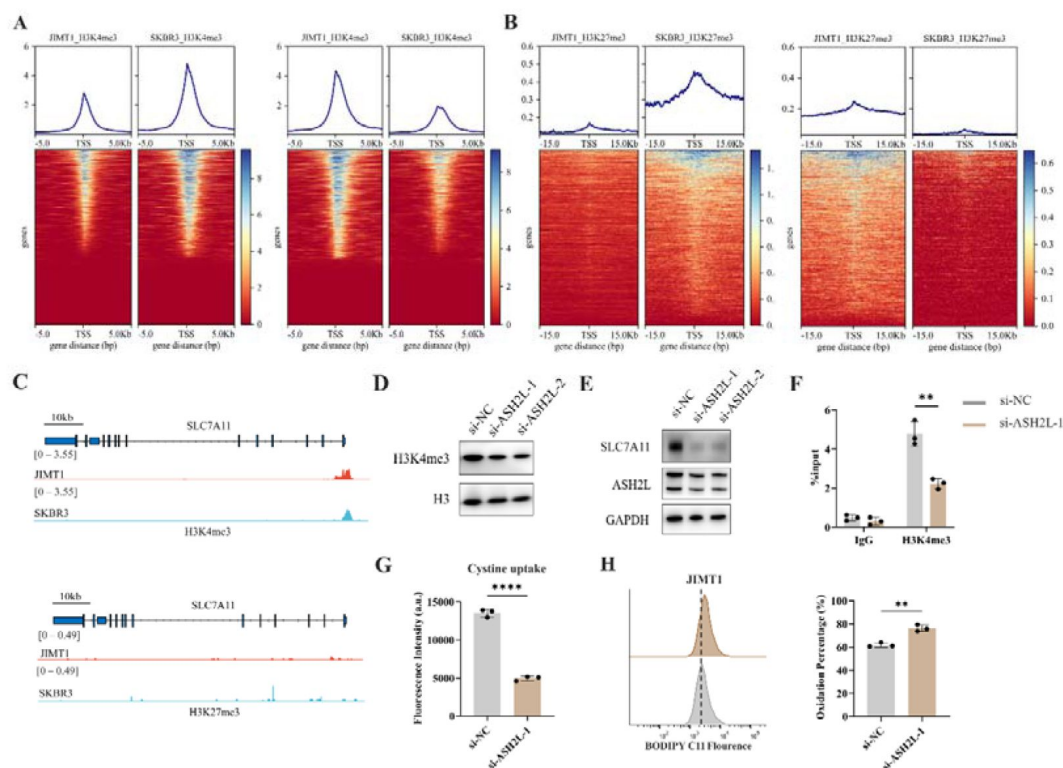


Figure 5.

Increased H3K4me3 modifies SLC7A11 transcription and regulated cystine uptake.

(A-B) Alterations of H3K4me3 (A) and H3K27me3 (B) peaks between JIMT1 and SKBR3. (C) Different abundance of H3K4me3 and H3K27me3 at SLC7A11 promoter regions in JIMT1 and SKBR3. (D) Decreased ASH2L inhibited total H3K4me3 expression. (E) Decreased ASH2L inhibited SLC7A11 expression. (F) Decreased ASH2L suppressed H3K4me3 expression at SLC7A11 promoter regions. (G) Decreased ASH2L suppressed cystine uptake ability. (H) Decreased ASH2L induced lipid peroxidation and ferroptosis. Significances were determined by two-tailed unpaired t-test (F-H). ns, $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

DNA methylation modifies SLC7A11 transcription and cysteine metabolism

Besides histone modification, DNA methylation is another crucial epigenetic modification to regulate genomic imprinting, modify transposable elements and modulate transcription, even in tumor progression(38 [↗](#)).

In general, JIMT1 cells featured a lower abundance of total 5-mC than SKBR3 cells (Figure S9A [↗](#)). By investigating DNA methylation landscape with WGBS, we identified 329419 CG-DMRs enriched in JIMT1 and 84634 CG-DMRs enriched in SKBR3 (Figure 6A [↗](#) and S9B [↗](#)). Among non-CG DMRs (CHG, CHH; in which H denotes A, C or T), 1604 CHG-DMRs and 3727 CHH-DMRs were upregulated in JIMT1, while 2187 CHG-DMRs and 5158 CHH-DMRs were upregulated in SKBR3 (Figure S9B [↗](#)).

Across whole genome, CG-DMRs mainly enriched in intron, intergenic regions, and exons (Figure S9C [↗](#)). Differentially methylated genes were related to several biological processes, such as neurotransmitter biosynthesis, postsynaptic transmission, cytoplasmic translation, and ribosomal activities (Figure S10A and S10B [↗](#)).

SLC7A11 is located at chromosome 4q28.3 and we found that, in comparison with SKBR3, the CpG methylation at SLC7A11 promoter region was decreased in JIMT1 (Figure 6B [↗](#)). To further verify the influence of altered promoter GC-DMR on SLC7A11 gene transcription, CRISPR/Cas9-based genomic editing tool dCas9-DNMT3A together with SLC7A11 specific promoter-targeting sgRNA sets were utilized to successfully increase the 5-mC level at SLC7A11 promoter regions (Figure 6C [↗](#)) in JIMT1 cells, which is confirmed by MeDIP-qPCR (Figure 6D [↗](#)). As a result, the expression of SLC7A11 and cysteine uptake were reduced (Figure 6E [↗](#) and 6F [↗](#)), and the following cellular lipid peroxidation and ferroptosis were stimulated in JIMT1 cells (Figure 6G [↗](#)).

These results revealed that increased DNA methylation at SLC7A11 promoter regions in JIMT1 might contribute to SLC7A11 overexpression. Precise modulation of 5-mC could suppress cysteine uptake and result in ferroptosis, which might provide novel methods to overcome trastuzumab resistance.

Discussion

Metabolic heterogeneity has been widely accepted as important characteristics in many cancer types, which are closely associated with cancer progression, distant metastasis, and even drug resistance(39 [↗](#), 40 [↗](#)), while targeting dysregulated metabolism could also be novel therapeutic strategies(41 [↗](#)).

By comparing the metabolomic data of cell lines and blood samples from patients, we realized that widespread metabolic reprogramming particular the alternation of cysteine metabolism was an important feature of primary trastuzumab resistance, extending our understanding of anti-tumor drug resistance (Figure 7 [↗](#)).

Cysteine is a crucial non-essential amino acid and mainly obtained from dietary, protein catabolism and methionine transsulfuration(42 [↗](#)). Due to oxidative extracellular environment, cystine, instead of cysteine, is normally intake by the major transporter SLC7A11 (also known as xCT)(43 [↗](#)), which also has important roles in protein synthesis, posttranslational modifications, and redox balance maintenance(42 [↗](#), 44 [↗](#)).

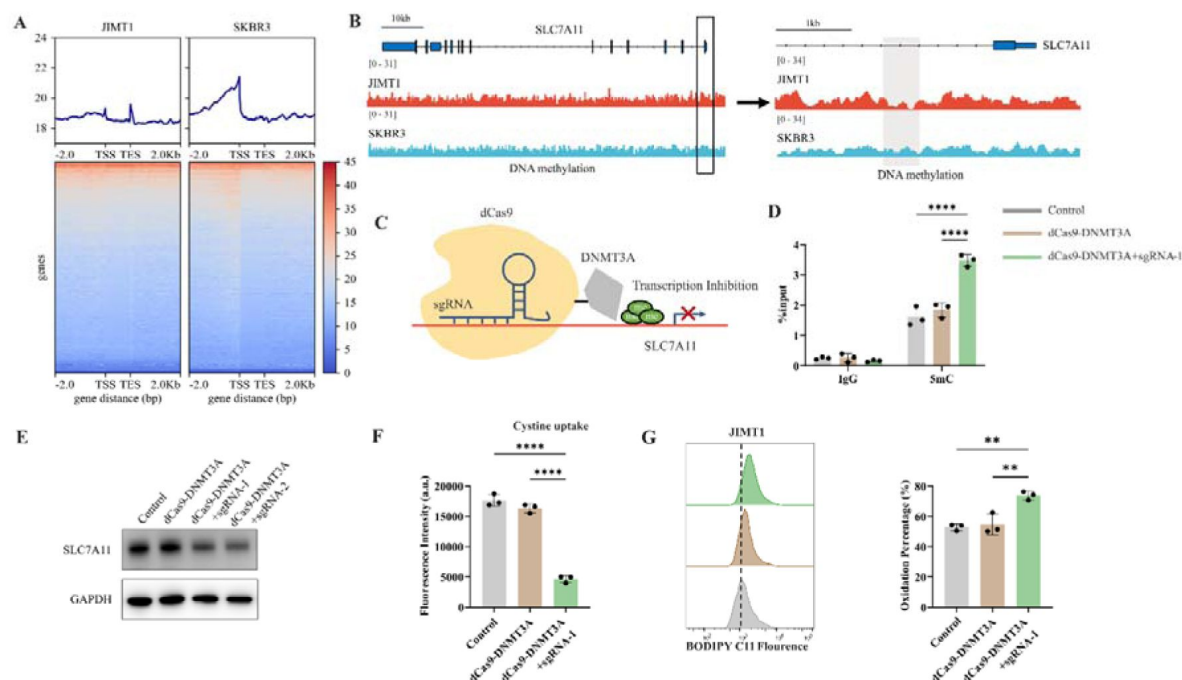


Figure 6.

Decreased 5-mC modifications on SLC7A11 promoter relate with enhanced cystine uptake.

(A) Alterations of 5-mC peaks between JIMT1 and SKBR3. (B) Different abundance of 5-mC at SLC7A11 promoter regions in JIMT1 and SKBR3. (C) Schematic outline of CRISPR-based epigenetic editing. dCas9-DNMT3A increased methylation of specific CpG islands in SLC7A11 promoter regions and inhibited gene transcription. (D) dCas9-DNMT3A increased 5-mC levels at SLC7A11 promoter regions. (E) dCas9-DNMT3A inhibited SLC7A11 expression. (F) dCas9-DNMT3A suppressed cystine uptake ability. (G) dCas9-DNMT3A induced lipid peroxidation and ferroptosis. Significances were determined by one-way ANOVA (D, F, G). ns, $P \geq 0.05$; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

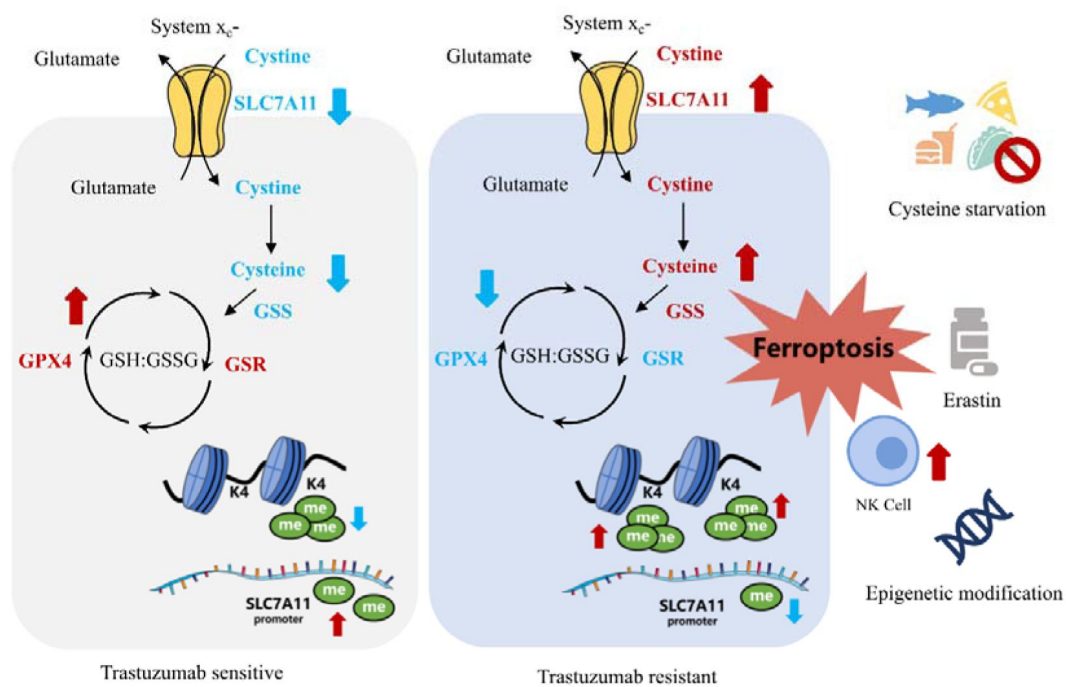


Figure 7.

Summary of targeting cystine metabolism in trastuzumab resistant HER2 positive breast cancer.

Meanwhile, biosynthesis of GSH by GSS is one important outlet of cellular cysteine, which helps maintaining oxidation-reduction balance and protecting membrane from oxidative damage including ferroptosis with GPX4 and GSR(45 [↗](#), 46 [↗](#)). The downregulation of both GPX4 and GSR and the upregulation of GSS, together with the higher intake of cysteine mediated by raised SLC7A11, could stimulate GSH consumption and GSSG accumulation in JIMT1 cells, resulting in a weaker reductive cytoplasmic environment and higher sensitivity to oxidative damage, including ferroptosis.

As an important rate-limiting metabolite of the GSH biosynthesis, cysteine modulates ferroptosis through the cyst(e)ine/GSH/GPX4 axis(47 [↗](#)). SLC7A11 inhibition, cysteine deprivation or targeting GPX4 have been reported to effectively enhance lipid peroxidation and induce ferroptosis in several cancer types(48 [↗](#)–51 [↗](#)). In our study, JIMT1 was much more sensitive to SLC7A11 and GPX4 inhibition, but tolerable to cysteine starvation, and this might result from increased methionine transsulfuration activity and γ -glutamyl-peptides synthesis, which compensated the deficiency of extracellular cysteine(52 [↗](#), 53 [↗](#)). Whether similar effects could be observed on GSS or GSR also need to be studied.

The inhibition of HER2 could result in impaired redox homeostasis and increased ROS(54 [↗](#)). Tumor cells might adapt to this metabolic stress by activating antioxidant signaling pathways and acquire recurrence and dormancy(54 [↗](#), 55 [↗](#)). Although JIMT1 is resistant to trastuzumab, the extended exposure of trastuzumab was found to decrease SLC7A11 expression. SLC7A11 overexpression and altered cysteine and glutathione metabolism might function as compensating ways for trastuzumab adaptation. The dual inhibition of cysteine metabolism and HER2 also enhanced ferroptosis both in vivo and in vitro. In a similar manner, the inhibition of glutamine metabolism and antioxidant transcription factor NRF2 could also suppress the proliferation of residual breast cancer after HER2 inhibition and restore intracellular GSH metabolism(54 [↗](#)). These results indicated that re-establishing redox homeostasis and modulating antioxidant metabolism might provide novel treatment options for trastuzumab primary resistant HER2 positive breast cancer.

Besides amino acids metabolism, lipid metabolism, especially polyunsaturated fatty acids (PUFAs) and relevant phospholipid metabolism, which are also key drivers of ferroptosis(56 [↗](#)), were identified dysregulated in both primary and acquired trastuzumab resistance of HER2 positive breast cancer(57 [↗](#)), demonstrating the possibility that targeting PUFA metabolism to induce ferroptosis might provide novel treatment options for overcoming trastuzumab resistance.

Not limited to cancer cells, altered cellular metabolism and oxidation-reduction balance can also affect activities of immune cells within TME(58 [↗](#)–60 [↗](#)). For example, lipid peroxidation stress might impair NK functions and immune surveillance. Metabolites, such as L-kynurenine, might enhance ferroptosis in NK cells through aryl hydrocarbon receptor and mediate immunosuppressive TME in gastric cancer(61 [↗](#)). Activation of NRF2 related antioxidant pathways could restore NK cell functions and trigger enhanced anti-tumor responses(62 [↗](#)). And The deprivation of cysteine was reported to promote the metabolic reprogramming of TME and influence T cell immune responses(63 [↗](#)). We observed that the combination of cysteine starvation and trastuzumab can increase tumor infiltrating NK cells but show limited effects on NK cell development and toxic functions, relevant mechanisms still need to be discussed. In addition, metabolites, cytokines, damage-associated molecular patterns (DAMPs) and other immunomodulatory signals released from cancer cells that undergo ferroptosis could function on immune cells nearby, stimulating or suppressing anti-tumor immune responses(64 [↗](#), 65 [↗](#)).

The emergence of drug resistance in cancers mainly depends on two scenarios: Darwinian selection and Lamarckian induction(66 [↗](#), 67 [↗](#)). During the evolution of breast cancer, different subtypes of breast cancer might originate from different development status of mammary stem cells, bipotent progenitors and differentiated cells(68 [↗](#), 69 [↗](#)). HER2 positive breast cancer was

reported to transform from luminal progenitor and mature luminal cells(70–72). The tumorigenesis of trastuzumab primary resistant HER2 positive breast cancer might favor the Darwinian selection, as a small population of pre-existing trastuzumab tolerant cancer cells might be enriched upon treatment.

To investigate genetic and epigenetic issues stimulating the evolution of primary trastuzumab resistant and sensitive breast cancer, we compared genomic variation, histone modification and DNA methylation status among JIMT1, SKBR3 and MCF10A (normal mammary epithelial cell).

During carcinogenesis, missense mutations, splicing mutations and in frame deletions were the most frequent events and MUC3A, MUC6, OR8U1 and ZNF717 were most frequent mutated genes (Figure S11A and S11B). Meanwhile, single nucleotide polymorphism (SNP) was the major variation type and the substitution between cytosine and thymine was the most common signature (Figure S12A-B and S13A-C).

Compared with MCF10A, JIMT1 instead of SKBR3 featured distinct single nucleotide variation (SNV) at the exon region of SLC7A11 (gain at Chr4:138183261 and loss at Chr4: 138219340) (Figure S14A-C).

SNVs located in exons have been reported to modulate gene transcription and function. For example, SNVs in BRCA1 exons encoding RING and BRCT domains might disrupt BRCA1 expression and function, therefore increasing breast cancer risks(73). While noncoding SNVs, such as rs9383590 located at ESR1 enhancer region, could interact with regulatory elements and activate oncogene transcription(74). The accumulation of SNVs was also associated with driver gene mutations and might promote breast cancer evolution in a cumulative manner(75). Whether different SNVs in SLC7A11 exon regions could contribute to SLC7A11 expression and breast cancer initiation still needs to be studied.

Epigenetic issues also demonstrated their impact on carcinogenesis. H3K4me3 instead of H3K27me3 peaks were shown more enrichment at SLC7A11 promoter regions in MCF10A cells (Figure S15A and S15B), while 5-mC levels at CpG islands among SLC7A11 promoter regions were also increased in MCF10A (Figure S15C), revealing the important proposal of histone modifications and DNA methylation on metabolic reprogramming during the evolution of HER2 positive breast cancer.

Although several studies have reported that SLC7A11 expression was modulated by NRF2, ATF4 and other transcription factors(43, 76, 77), it remains unclear whether its expression is subject to a broader genome-wide or histone-level regulation(78). The theory that epigenetic modifications contribute to specific transcription regulation, cancer cell development and metabolic reprogramming, to modulate treatment response and drug resistance, has been widely accepted and proved(79). Our study tried to explore the relationship between the altered cysteine metabolism signature and epigenetic modifications in trastuzumab resistant HER2 positive breast cancer. Elucidating epigenetic mechanisms would help better understand trastuzumab resistance formation and provide novel therapeutic opportunities.

Several studies have described the regulatory roles of histone modifications on oncogene transcription(79, 80). MLL mutations were identified among various cancers and widespread shortening H3K4me3 peaks at tumor suppressor genes might contribute to tumor initiation(81, 82) and H3K27me3 could modulate chemotherapy tolerance in triple negative breast cancer (TNBC)(83). Moreover, the combination of chemotherapy and H3K27me3 demethylation inhibitor could prevent the proliferation of persister cells(83). Targeting KDM6A could inhibit JIMT1 tumor growth via decreasing tRNA transcription(84). As we confirmed, reducing

H3K4me3 can decrease ASH2L, SLC7A11 expression and cystine uptake, leading to enhanced lipid peroxidation and ferroptosis, modulating histone modification can be a promising strategy in cancer treatment.

Compared with normal tissues, almost all cancers feature abnormal DNA hypomethylation and local hypermethylation at specific CpG islands(81 [↗](#), 85 [↗](#)). Dysregulated DNA methylation contributes to breast cancer tumorigenesis and drug resistance formation(86 [↗](#), 87 [↗](#)). For instance, trastuzumab resistant HER2 positive breast cancer was found associated with downregulated TGFBI, CXCL2 and SLC38A1, which was silenced by DNA hypermethylation(88 [↗](#)). Aberrant DNMT1, DNMT3A and DNMT3B expression were correlated with endocrine therapy resistance and predicted worsen pathologic status and prognosis(89 [↗](#)). By utilizing CRISPR interference, we verified the effect of 5-mC at SLC7A11 promoter on gene expression and downstream cysteine metabolism. What's more, since increased H3K4me3 might interact with DNMT3A-DNMT3L(90 [↗](#)), the crosstalk between histone modification and DNA methylation may have collaborative effects on cysteine metabolism in trastuzumab primary resistant breast cancer.

In conclusion, our study revealed that trastuzumab primary resistant HER2 positive breast cancer featured distinct cysteine metabolism, which was driven by altered H3K4me3 and DNA methylation. Targeting cysteine metabolism could provide novel treatment concepts for overcoming trastuzumab resistance.

Data availability

Further information and requests for resources could be directed to corresponding authors.

Acknowledgements

We would like to thank the Core Facility of the First Affiliated Hospital of Nanjing Medical University for its help in the detection of experimental samples.

Additional information

Funding

This project was supported by National Natural Science Foundation of China (81972484 and 82272667 to Yongmei Yin; 82203488 to Ningjun Duan), High-level Innovation Team of Nanjing Medical University Program (JX102GSP201727), The Collaborative Innovation Center for Tumor Individualization Program (JX21817902/008) and Beijing Xisike Clinical Oncology Research Foundation (Y-2019AZZD-00680).

Author Contributions

Y.M.Y. and N.J.D. conceived the project. Y.J.H., N.J.D., C.X.S., F.Y., Y.T.S., S.H.Z., Y.L. and J.G. performed the experiments. M.T. and Q.L. helped collect clinical samples. Y.J.H. and N.J.D. conducted data analysis and wrote the manuscript. N.J.D., X.H., Z.Y.F., W.L. and Y.M.Y. supervised the study. All authors reviewed the manuscript.

Abbreviations

4-HNE, 4-hydroxynonenal; 5-mC, 5-methylcytosine; ADCC, antibody dependent cellular cytotoxicity; DAMPs, damage-associated molecular patterns; DMR, differentially methylated regions; GABA, 4-aminobutyric acid; GSH, glutathione; GSR, glutathione reductase; GSS, glutathione synthetase; GSSG, glutathione disulfide; HER2, human epidermal growth factor receptor 2; MDA, Malondialdehyde; pCR, pathological complete response; PUFAs, polyunsaturated fatty acids; PUMA, p53 upregulated modulator of apoptosis; sgRNA, small guide RNA; siRNA, short interfering RNA; SNP, single nucleotide polymorphism; SNV, single nucleotide variation; TCA, tricarboxylic acid; TME, tumor microenvironment; TNBC, triple negative breast cancer; WGBS, whole-genome bisulfite sequencing.

Additional files

Figures S1-S15 and Tables S1-S3 [🔗](#)

References

1. Giaquinto AN, Sung H, Miller KD, et al. (2022) **Breast Cancer Statistics, 2022** *CA Cancer J Clin* **72**:524–41
2. Harbeck N, Penault-Llorca F, Cortes J, et al. (2019) **Breast cancer** *Nat Rev Dis Primers* **5**:66
3. Swain SM, Shastry M, Hamilton E (2023) **Targeting HER2-positive breast cancer: advances and future directions** *Nat Rev Drug Discov* **22**:101–26
4. Marra A, Chandarlapaty S, Modi S (2024) **Management of patients with advanced-stage HER2-positive breast cancer: current evidence and future perspectives** *Nat Rev Clin Oncol* **21**:185–202
5. Cossetti RJ, Tyldesley SK, Speers CH, et al. (2015) **Comparison of breast cancer recurrence and outcome patterns between patients treated from 1986 to 1992 and from 2004 to 2008** *J Clin Oncol* **33**:65–73
6. Xu X, De Angelis C, Burke KA, et al. (2017) **HER2 Reactivation through Acquisition of the HER2 L755S Mutation as a Mechanism of Acquired Resistance to HER2-targeted Therapy in HER2(+) Breast Cancer** *Clin Cancer Res* **23**:5123–34
7. Arribas J, Baselga J, Pedersen K, et al. (2011) **p95HER2 and breast cancer** *Cancer Res* **71**:1515–9
8. Baselga J, Cortes J, Im SA, et al. (2014) **Biomarker analyses in CLEOPATRA: a phase III, placebo-controlled study of pertuzumab in human epidermal growth factor receptor 2-positive, first-line metastatic breast cancer** *J Clin Oncol* **32**:3753–61
9. Marchio C, Annaratone L, Marques A, et al. (2021) **Evolving concepts in HER2 evaluation in breast cancer: Heterogeneity HER2-low carcinomas and beyond** *Semin Cancer Biol* **72**:123–35
10. Hanahan D (2022) **Hallmarks of Cancer: New Dimensions** *Cancer Discov* **12**:31–46
11. Martinez-Reyes I, Chandel NS (2021) **Cancer metabolism: looking forward** *Nat Rev Cancer* **21**:669–80
12. Heiden MG Vander, DeBerardinis RJ (2017) **Understanding the Intersections between Metabolism and Cancer Biology** *Cell* **168**:657–69
13. Kao KC, Vilbois S, Tsai CH, et al. (2022) **Metabolic communication in the tumour-immune microenvironment** *Nat Cell Biol* **24**:1574–83
14. Gong Y, Ji P, Yang YS, et al. (2021) **Metabolic-Pathway-Based Subtyping of Triple-Negative Breast Cancer Reveals Potential Therapeutic Targets** *Cell Metab* **33**:51–64
15. Xiao Y, Ma D, Yang YS, et al. (2022) **Comprehensive metabolomics expands precision medicine for triple-negative breast cancer** *Cell Res* **32**:477–90

16. Miller KD, O'Connor S, Pniewski KA, et al. (2023) **Acetate acts as a metabolic immunomodulator by bolstering T-cell effector function and potentiating antitumor immunity in breast cancer** *Nat Cancer* **4**:1491–507
17. Xiao Y, Yu TJ, Xu Y, et al. (2023) **Emerging therapies in cancer metabolism** *Cell Metab* **35**:1283–303
18. Timp W, Feinberg AP (2013) **Cancer as a dysregulated epigenome allowing cellular growth advantage at the expense of the host** *Nat Rev Cancer* **13**:497–510
19. Feinberg AP, Koldobskiy MA, Gondor A (2016) **Epigenetic modulators, modifiers and mediators in cancer aetiology and progression** *Nat Rev Genet* **17**:284–99
20. Brown R, Curry E, Magnani L, et al. (2014) **Poised epigenetic states and acquired drug resistance in cancer** *Nat Rev Cancer* **14**:747–53
21. Langille E, Al-Zahrani KN, Ma Z, et al. (2022) **Loss of Epigenetic Regulation Disrupts Lineage Integrity, Induces Aberrant Alveogenesis, and Promotes Breast Cancer** *Cancer Discov* **12**:2930–53
22. Garcia-Martinez L, Zhang Y, Nakata Y, et al. (2021) **Epigenetic mechanisms in breast cancer therapy and resistance** *Nat Commun* **12**:1786
23. Wong H, Leung R, Kwong A, et al. (2011) **Integrating molecular mechanisms and clinical evidence in the management of trastuzumab resistant or refractory HER-2(+) metastatic breast cancer** *Oncologist* **16**:1535–46
24. Doench JG, Fusi N, Sullender M, et al. (2016) **Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9** *Nat Biotechnol* **34**:184–91
25. Sanson KR, Hanna RE, Hegde M, et al. (2018) **Optimized libraries for CRISPR-Cas9 genetic screens with multiple modalities** *Nat Commun* **9**:5416
26. O'Geen H, Ren C, Nicolet CM, et al. (2017) **dCas9-based epigenome editing suggests acquisition of histone methylation is not sufficient for target gene repression** *Nucleic Acids Res* **45**:9901–16
27. Ewald JD, Zhou G, Lu Y, et al. (2024) **Web-based multi-omics integration using the Analyst software suite** *Nat Protoc*
28. Ramirez F, Ryan DP, Gruning B, et al. (2016) **deepTools2: a next generation web server for deep-sequencing data analysis** *Nucleic Acids Res* **44**:W160–5
29. Krueger F, Andrews SR (2011) **Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications** *Bioinformatics* **27**:1571–2
30. Clark AS, Yau C, Wolf DM, et al. (2021) **Neoadjuvant T-DM1/pertuzumab and paclitaxel/trastuzumab/pertuzumab for HER2(+) breast cancer in the adaptively randomized I-SPY2 trial** *Nat Commun* **12**:6428
31. Dixon SJ, Olzmann JA (2024) **The cell biology of ferroptosis** *Nat Rev Mol Cell Biol*
32. Ye L, Wen X, Qin J, et al. (2024) **Metabolism-regulated ferroptosis in cancer progression and therapy** *Cell Death Dis* **15**:196

33. Musolino A, Gradishar WJ, Rugo HS, et al. (2022) **Role of Fcγ receptors in HER2-targeted breast cancer therapy** *J Immunother Cancer* **10**
34. Lee YS, Kalimuthu K, Park YS, et al. (2020) **BAX-dependent mitochondrial pathway mediates the crosstalk between ferroptosis and apoptosis** *Apoptosis* **25**:625–31
35. Jin ML, Jeong KW (2023) **Histone modifications in drug-resistant cancers: From a cancer stem cell and immune evasion perspective** *Exp Mol Med* **55**:1333–47
36. Millan-Zambrano G, Burton A, Bannister AJ, et al. (2022) **Histone post-translational modifications - cause and consequence of genome function** *Nat Rev Genet* **23**:563–80
37. Piunti A, Shilatifard A (2016) **Epigenetic balance of gene expression by Polycomb and COMPASS families** *Science* **352**:aad9780
38. Mattei AL, Bailly N, Meissner A (2022) **DNA methylation: a historical perspective** *Trends Genet* **38**:676–707
39. Faubert B, Solmonson A, DeBerardinis RJ (2020) **Metabolic reprogramming and cancer progression** *Science* :368
40. Kim J, DeBerardinis RJ (2019) **Mechanisms and Implications of Metabolic Heterogeneity in Cancer** *Cell Metab* **30**:434–46
41. Stine ZE, Schug ZT, Salvino JM, et al. (2022) **Targeting cancer metabolism in the era of precision oncology** *Nat Rev Drug Discov* **21**:141–62
42. Bonifacio VDB, Pereira SA, Serpa J, et al. (2021) **Cysteine metabolic circuitries: druggable targets in cancer** *Br J Cancer* **124**:862–79
43. Koppula P, Zhuang L, Gan B (2021) **Cystine transporter SLC7A11/xCT in cancer: ferroptosis, nutrient dependency, and cancer therapy** *Protein Cell* **12**:599–620
44. Lennicke C, Cocheme HM (2021) **Redox metabolism: ROS as specific molecular regulators of cell signaling and function** *Mol Cell* **81**:3691–707
45. Sies H, Mailloux RJ, Jakob U (2024) **Fundamentals of redox regulation in biology** *Nat Rev Mol Cell Biol*
46. Wu K, El Zowalaty AE, Sayin VI, et al. (2024) **The pleiotropic functions of reactive oxygen species in cancer** *Nat Cancer* **5**:384–99
47. Stockwell BR (2022) **Ferroptosis turns 10: Emerging mechanisms, physiological functions, and therapeutic applications** *Cell* **185**:2401–21
48. Dixon SJ, Lemberg KM, Lamprecht MR, et al. (2012) **Ferroptosis: an iron-dependent form of nonapoptotic cell death** *Cell* **149**:1060–72
49. Badgley MA, Kremer DM, Maurer HC, et al. (2020) **Cysteine depletion induces pancreatic tumor ferroptosis in mice** *Science* **368**:85–9
50. Shi Z, Naowarajna N, Pan Z, et al. (2021) **Multifaceted mechanisms mediating cystine starvation-induced ferroptosis** *Nat Commun* **12**:4792

51. Yang F, Xiao Y, Ding JH, et al. (2023) **Ferroptosis heterogeneity in triple-negative breast cancer reveals an innovative immunotherapy combination strategy** *Cell Metab* **35**:84–100
52. Zhang HF, Klein Geltink RI, Parker SJ, et al. (2022) **Transsulfuration, minor player or crucial for cysteine homeostasis in cancer** *Trends Cell Biol* **32**:800–14
53. Kang YP, Mockabee-Macias A, Jiang C, et al. (2021) **Non-canonical Glutamate-Cysteine Ligase Activity Protects against Ferroptosis** *Cell Metab* **33**:174–89
54. Fox DB, Garcia NMG, McKinney BJ, et al. (2020) **NRF2 activation promotes the recurrence of dormant tumour cells through regulation of redox and nucleotide metabolism** *Nat Metab* **2**:318–34
55. Parida PK, Marquez-Palencia M, Nair V, et al. (2022) **Metabolic diversity within breast cancer brain-tropic cells determines metastatic fitness** *Cell Metab* **34**:90–105
56. Pope LE, Dixon SJ (2023) **Regulation of ferroptosis by lipid metabolism** *Trends Cell Biol* **33**:1077–87
57. Duan N, Hua Y, Yan X, et al. (2024) **Unveiling Alterations of Epigenetic Modifications and Chromatin Architecture Leading to Lipid Metabolic Reprogramming during the Evolutionary Trastuzumab Adaptation of HER2-Positive Breast Cancer** *Adv Sci (Weinh)* :e2309424
58. De Martino M, Rathmell JC, Galluzzi L, et al. (2024) **Cancer cell metabolism and antitumour immunity** *Nat Rev Immunol*
59. Leone RD, Powell JD (2020) **Metabolism of immune cells in cancer** *Nat Rev Cancer* **20**:516–31
60. Chen X, Kang R, Kroemer G, et al. (2021) **Broadening horizons: the role of ferroptosis in cancer** *Nat Rev Clin Oncol* **18**:280–96
61. Cui JX, Xu XH, He T, et al. (2023) **L-kynurenine induces NK cell loss in gastric cancer microenvironment via promoting ferroptosis** *J Exp Clin Cancer Res* **42**:52
62. Poznanski SM, Singh K, Ritchie TM, et al. (2021) **Metabolic flexibility determines human NK cell functional fate in the tumor microenvironment** *Cell Metab* **33**:1205–20
63. Han C, Ge M, Xing P, et al. (2024) **Cystine deprivation triggers CD36-mediated ferroptosis and dysfunction of tumor infiltrating CD8(+) T cells** *Cell Death Dis* **15**:145
64. Dang Q, Sun Z, Wang Y, et al. (2022) **Ferroptosis: a double-edged sword mediating immune tolerance of cancer** *Cell Death Dis* **13**:925
65. Zhou Q, Meng Y, Li D, et al. (2024) **Ferroptosis in cancer: From molecular mechanisms to therapeutic strategies** *Signal Transduct Target Ther* **9**:55
66. Boumahdi S, de Sauvage FJ (2020) **The great escape: tumour cell plasticity in resistance to targeted therapy** *Nat Rev Drug Discov* **19**:39–56
67. Gatenby RA, Brown JS (2020) **Integrating evolutionary dynamics into cancer therapy** *Nat Rev Clin Oncol* **17**:675–86
68. Polyak K (2007) **Breast cancer: origins and evolution** *J Clin Invest* **117**:3155–63

69. Chaffer CL, Weinberg RA (2010) **Cancer cell of origin: spotlight on luminal progenitors** *Cell Stem Cell* **7**:271–2
70. Lim E, Vaillant F, Wu D, et al. (2009) **Aberrant luminal progenitors as the candidate target population for basal tumor development in BRCA1 mutation carriers** *Nat Med* **15**:907–13
71. Prat A, Perou CM (2009) **Mammary development meets cancer genomics** *Nat Med* **15**:842–4
72. Tharmapalan P, Mahendralingam M, Berman HK, et al. (2019) **Mammary stem cells and progenitors: targeting the roots of breast cancer for prevention** *EMBO J* **38**:e100852
73. Findlay GM, Daza RM, Martin B, et al. (2018) **Accurate classification of BRCA1 variants with saturation genome editing** *Nature* **562**:217–22
74. Bailey SD, Desai K, Kron KJ, et al. (2016) **Noncoding somatic and inherited single-nucleotide variants converge to promote ESR1 expression in breast cancer** *Nat Genet* **48**:1260–6
75. Nishimura T, Kakiuchi N, Yoshida K, et al. (2023) **Evolutionary histories of breast cancer and related clones** *Nature* **620**:607–14
76. Anandhan A, Dodson M, Schmidlin CJ, et al. (2020) **Breakdown of an Ironclad Defense System: The Critical Role of NRF2 in Mediating Ferroptosis** *Cell Chem Biol* **27**:436–47
77. Bai X, Ni J, Beretov J, et al. (2021) **Activation of the eIF2alpha/ATF4 axis drives triple-negative breast cancer radioresistance by promoting glutathione biosynthesis** *Redox Biol* **43**
78. Wang Y, Hu J, Wu S, et al. (2023) **Targeting epigenetic and posttranslational modifications regulating ferroptosis for the treatment of diseases** *Signal Transduct Target Ther* **8**:449
79. Wang N, Ma T, Yu B (2023) **Targeting epigenetic regulators to overcome drug resistance in cancers** *Signal Transduct Target Ther* **8**:69
80. Sun L, Zhang H, Gao P (2022) **Metabolic reprogramming and epigenetic modifications on the path to cancer** *Protein Cell* **13**:877–919
81. Zhao S, Allis CD, Wang GG (2021) **The language of chromatin modification in human cancers** *Nat Rev Cancer* **21**:413–30
82. Chen K, Chen Z, Wu D, et al. (2015) **Broad H3K4me3 is associated with increased transcription elongation and enhancer activity at tumor-suppressor genes** *Nat Genet* **47**:1149–57
83. Marsolier J, Prompsy P, Durand A, et al. (2022) **H3K27me3 conditions chemotolerance in triple-negative breast cancer** *Nat Genet* **54**:459–68
84. Duan N, Hua Y, Yan X, et al. (2024) **An Imbalance in Histone Modifiers Induces tRNA-Cys-GCA Overexpression and tRF-27 Accumulation by Attenuating Promoter H3K27me3 in Primary Trastuzumab-Resistant Breast Cancer** *Cancers (Basel)* **16**
85. Nishiyama A, Nakanishi M (2021) **Navigating the DNA methylation landscape of cancer** *Trends Genet* **37**:1012–27

86. Sher G, Salman NA, Khan AQ, et al. (2022) **Epigenetic and breast cancer therapy: Promising diagnostic and therapeutic applications** *Semin Cancer Biol* **83**:152–65
87. Sukocheva OA, Lukina E, Friedemann M, et al. (2022) **The crucial role of epigenetic regulation in breast cancer anti-estrogen resistance: Current findings and future perspectives** *Semin Cancer Biol* **82**:35–59
88. Palomeras S, Diaz-Lagares A, Vinas G, et al. (2019) **Epigenetic silencing of TGFBI confers resistance to trastuzumab in human breast cancer** *Breast Cancer Res* **21**:79
89. Jahangiri R, Jamialahmadi K, Gharib M, et al. (2019) **Expression and clinicopathological significance of DNA methyltransferase 1, 3A and 3B in tamoxifen-treated breast cancer patients** *Gene* **685**:24–31
90. Du J, Johnson LM, Jacobsen SE, et al. (2015) **DNA methylation pathways and their crosstalk with histone methylation** *Nat Rev Mol Cell Biol* **16**:519–32

Author information

Yijia Hua[#]

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China, Gusu School, Suzhou Municipal Hospital, The Affiliated Suzhou Hospital of Nanjing Medical University, Suzhou, China
ORCID iD: [0009-0006-4323-8715](https://orcid.org/0009-0006-4323-8715)

[#]These authors contributed equally to this work.

Ningjun Duan[#]

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China
ORCID iD: [0000-0001-6316-3796](https://orcid.org/0000-0001-6316-3796)

For correspondence: ningjun_official@outlook.com

[#]These authors contributed equally to this work.

Chunxiao Sun

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Fan Yang

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Min Tian

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Yanting Sun

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Shuhan Zhao

Department of General Surgery, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Jue Gong

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Qian Liu

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Xiang Huang

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Yan Liang

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Ziyi Fu

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Wei Li

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China

Yongmei Yin

Department of Oncology, The First Affiliated Hospital of Nanjing Medical University, Nanjing, China, Gusu School, Suzhou Municipal Hospital, The Affiliated Suzhou Hospital of Nanjing Medical University, Suzhou, China, Jiangsu Key Lab of Cancer Biomarkers, Prevention and Treatment, Collaborative Innovation Center for Personalized Cancer Medicine, Nanjing Medical University, Nanjing, China

For correspondence: ymyin@njmu.edu.cn

Editors

Reviewing Editor

Yongliang Yang

Dalian University of Technology, Dalian, China

Senior Editor

Caigang Liu

Shengjing Hospital of China Medical University, Shenyang, China

Reviewer #1 (Public review):

Summary:

Hua et al show how targeting amino acid metabolism can overcome Trastuzumab resistance in HER2+ breast cancer.

Strengths:

The authors used metabolomics, transcriptomics and epigenomics approaches in vitro and in preclinical models to demonstrate how trastuzumab resistant cells utilize cysteine metabolism.

Weaknesses:

However, there are some key aspects that needs to be addressed.

Major:

(1) Patient Samples for Transcriptomic Analysis: It is unclear from the text whether tumor tissues or blood samples were used for the transcriptomic analysis. This distinction is crucial, as these two sample types would yield vastly different inferences. The authors should clarify the source of these samples.

(2) The study only tested one trastuzumab-resistant and one trastuzumab-sensitive cell lines. It is unclear whether these findings are applicable to other HER2-positive tumor cell lines, such as HCC1954. The authors should validate their results in additional cell lines to strengthen their conclusions.

(3) Relevance to Metastatic Disease: Trastuzumab resistance often arises in patients during disease recurrence, which is frequently associated with metastasis. However, the mouse experiments described in this paper were conducted only in the primary tumors. This article will have more impact if the authors could demonstrate that the combination of Erastin or cysteine starvation with trastuzumab can also improve outcomes in metastasis models.

Minor:

(1) The figures lack information about the specific statistical tests used. Including this information is essential to show the robustness of the results.

(2) Figure 3K Interpretation: The significance asterisks in Figure 3K do not specify the comparison being made. Are they relative to the DMSO control? This should be clarified.

Comments on revisions:

While the authors acknowledge the limitation of using only a single trastuzumab resistant/sensitive pair, simply stating that additional cell lines will be tested in future work is simply inadequate. The biological heterogeneity of HER2-positive breast cancer demands validation in at least one independent resistant model (e.g., HCC1954 or BT 474R) alongside its parental counterpart. Without demonstrating that SLC7A11 upregulation, cysteine dependency, and sensitivity to Erastin plus trastuzumab extend beyond the original cell line pair, the generalizability and translational relevance of the findings remain uncertain. The authors need to perform and report key functional results (cell viability, apoptosis, and SLC7A11 expression) in an additional resistant and sensitive HER2-positive cell line before this manuscript can be considered robust.

<https://doi.org/10.7554/eLife.103953.2.sa2>

Reviewer #2 (Public review):

In this manuscript, Hua et al. proposed SLC7A11, a protein facilitating cellular cystine uptake, as a potential target for the treatment of trastuzumab resistant HER2 positive breast cancer. If this claim holds true, the finding would be of significance and might be translated to clinical practice. Nevertheless, this reviewer finds that the conclusion was insufficiently supported by the data.

Notably, most of the data (Figures 2-6) were based on two cell lines - JIMT1 as a representative of trastuzumab resistant cell line, and SKBR3 as a representative of trastuzumab sensitive cell line. As such, these findings could be cell line specific while irrelevant to trastuzumab sensitivity at all. Furthermore, the authors' claim of ferroptosis induction is primarily based on lipid peroxidation assays (Figure 3). The rescuing effects of ferroptosis inhibitors on cell viability were missing. The xenograft experiments were also suspicious (Figure 4). Systemic cysteine starvation is known to cause adverse effects, including liver necrosis, and the compound (i.e., erastin) used by the authors is not suitable for in vivo experiments due to low solubility and low metabolic stability. Finally, the authors focus on epigenetic regulations (Figures 5 & 6) without first investigating well-established transcription factors, such as NRF2 and ATF4, which are known to regulate SLC7A11.

To sum up, this reviewer finds that the most valuable data in this manuscript is perhaps Figure 1, which provides unbiased information concerning the metabolic patterns in trastuzumab sensitive and primary resistant HER2 positive breast cancer patients.

Comments on revisions:

- (1) Figure 3: The unit of concentration should be " μM ". " μm " means micrometer.
- (2) Figure S5: Ferroptosis inhibitors should be used in cell viability assays to exclude the off-target effect of RSL3 and erastin. Note that erastin also targets VDAC, while RSL3 may inhibit other selenoproteins at high concentrations. Cell viability assays are critical for demonstrating ferroptosis and should be included in the main figure rather than relegated to the supplemental materials.
- (3) Figure 4B & 4C: the data of "H" group and "Erastin" group are inconsistent. In panel B, the tumor size in the "H" group appears smaller than in the "Erastin" group, while in panel C, the opposite trend is observed.
- (4) The catalog numbers for the cystine/cysteine-deficient DMEM (from BIOTREE) and diet (from Xietong Bio) should be provided. This information is essential for readers to identify and verify the specific products used in the study.

<https://doi.org/10.7554/eLife.103953.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

Hua et al show how targeting amino acid metabolism can overcome Trastuzumab resistance in HER2+ breast cancer.

Strengths:

The authors used metabolomics, transcriptomics and epigenomics approaches in vitro and in preclinical models to demonstrate how trastuzumab-resistant cells utilize cysteine metabolism.

Thank you for your valuable comments. We would like to extend our appreciation for your efforts. Your constructive suggestion would help improve our research.

Weaknesses:

However, there are some key aspects that needs to be addressed.

Major:

(1) Patient Samples for Transcriptomic Analysis: It is unclear from the text whether tumor tissues or blood samples were used for the transcriptomic analysis. This distinction is crucial, as these two sample types would yield vastly different inferences. The authors should clarify the source of these samples.

Thank you for your valuable comments. In the transcriptomic analysis, we included the data of HER2 positive breast cancer patients who received trastuzumab in I-SPY2 trial (GSE181574). Tumor tissues were used in this dataset. We highlighted the usage of “pre-treatment breast cancer tumors” in Line 309 and included the overview of transcriptomic data analysis in I-SPY2 trial in Figure S1F.

(2) The study only tested one trastuzumab-resistant and one trastuzumab-sensitive cell line. It is unclear whether these findings are applicable to other HER2-positive tumor cell lines, such as HCC1954. The authors should validate their results in additional cell lines to strengthen their conclusions.

Thank you for your valuable comments. We agree with your opinion, and the exploration of multiple cell lines would make our research findings more comprehensive. This is a limitation of our study, and we would continue to improve our design and methods in future experiments.

(3) Relevance to Metastatic Disease: Trastuzumab resistance often arises in patients during disease recurrence, which is frequently associated with metastasis. However, the mouse experiments described in this paper were conducted only in the primary tumors. This article would have more impact if the authors could demonstrate that the combination of Erastin or cysteine starvation with trastuzumab can also improve outcomes in metastasis models.

Thank you for your valuable comments. We agree with your suggestions. The exploration of metastatic disease would make our research more meaningful and help better address clinical key issues. In our future studies, we will continue to investigate the association between the invasive and metastatic capabilities of trastuzumab resistant HER2 positive breast cancer and cysteine metabolism.

Minor:

(1) The figures lack information about the specific statistical tests used. Including this information is essential to show the robustness of the results.

Thank you for your valuable comments. We added statistical information in our figure legends, including Line 849-850, Line 865-867, Line 881-882, Line 898-900, Line 910-911 and Line 923-924.

(2) Figure 3K Interpretation: The significance asterisks in Figure 3K do not specify the comparison being made. Are they relative to the DMSO control? This should be clarified.

Thank you for your valuable comments. We have modified this figure to demonstrate it more clearly. In Figure 3K, the significance was determined by one-way ANOVA and the comparison presented was relative to the DMSO control. It was indicated that the combination of erastin or cysteine starvation and trastuzumab could increase lipid peroxidation, although trastuzumab monotherapy did not induce ferroptosis.

Additionally, the combination of erastin and trastuzumab could result in more lipid peroxidation than erastin alone. Similar results were also found in the combination of cysteine starvation and trastuzumab. These results showed that targeting cysteine metabolism plus trastuzumab could have synergic effects to induce ferroptosis in trastuzumab resistant HER2 positive breast cancer.

Reviewer #2 (Public review):

In this manuscript, Hua et al. proposed SLC7A11, a protein facilitating cellular cystine uptake, as a potential target for the treatment of trastuzumab-resistant HER2-positive breast cancer. If this claim holds true, the finding would be of significance and might be translated to clinical practice. Nevertheless, this reviewer finds that the conclusion was poorly supported by the data.

Notably, most of the data (Figures 2-6) were based on two cell lines - JIMT1 as a representative of trastuzumab-resistant cell line, and SKBR3 as a representative of trastuzumab sensitive cell line. As such, these findings could be cell-line specific while irrelevant to trastuzumab sensitivity at all. Furthermore, the authors claimed ferroptosis simply based on lipid peroxidation (Figure 3). Cell viability was not determined, and the rescuing effects of ferroptosis inhibitors were missing. The xenograft experiments were also suspicious (Figure 4). The description of how cysteine starvation was performed on xenograft tumors was lacking, and the compound (i.e., erastin) used by the authors is not suitable for in vivo experiments due to low solubility and low metabolic stability. Finally, it is confusing why the authors focused on epigenetic regulations (Figures 5 & 6), without measuring major transcription factors (e.g., NRF2, ATF4) which are known to regulate SLC7A11.

To sum up, this reviewer finds that the most valuable data in this manuscript is perhaps Figure 1, which provides unbiased information concerning the metabolic patterns in trastuzumab-sensitive and primary resistant HER2-positive breast cancer patients.

Thank you for your valuable comments. We agree with your suggestions. Your feedback would help enhance the quality of our research.

(1) Our research was mainly conducted in JIMT1 (trastuzumab resistant) and SKBR3 (trastuzumab sensitive), and this is a limitation of our study. The experimental validation using different cell lines will make our research findings more persuasive. In our future research, we will continuously optimize experimental design and methods to make our findings more comprehensive.

(2) The detection of ferroptosis in our research was mainly performed by evaluating the lipid peroxidation. Experiments measuring cell viability and rescuing effects would help provide

more evidence.

We utilized CCK8 tests to compare cell viabilities of JIMT1 and SKBR3 in different erastin and RSL3 concentrations, as well as different exposure time of cysteine starvation. It was shown that JIMT1 was more sensitive to erastin and RSL3, but tolerant to cysteine starvation, which was consistent with the previous lipid peroxidation tests. This data was included in Figure S5C-E. We added the description in Line 375-379.

In addition, we also performed experiments to explore the rescuing effects of ferroptosis inhibitor Fer-1. It was indicated that Fer-1 could suppress the lipid peroxidation resulted from erastin, RSL3 and cysteine starvation in both JIMT1 and SKBR3. This provided more evidence that cysteine metabolism played a vital role in modulating HER2 positive breast cancer ferroptosis. This data was included in Figure S5G and S5H. We added the description to Line 387-391.

(3) In xenograft experiments, the cysteine starvation was performed by feeding cystine/cysteine-deficient diet (Xietong Bio). We added details of this diet on Line 236-237 in Methods.

We agree with your opinion on the role of erastin in experiments in vivo. We have tried to optimize drug dissolution and other conditions by referring to previous relevant literature. We would continue to improve our experimental design and methods.

(4) Epigenetic modifications have been recognized as crucial factors in drug resistance formation. An increasing number of studies have emphasized the importance of epigenetic changes in regulating the abnormal expression of oncogenes and tumor suppressor genes related to drug resistance. Currently, the role of epigenetic changes in the development of trastuzumab resistance in HER2 positive breast cancer is still in exploration. We tried to investigate the dysregulation of histone modifications and DNA methylation in trastuzumab resistant HER2 positive breast cancer. Our findings indicated that targeting H3K4me3 and DNA methylation could decrease SLC7A11 expression and induce ferroptosis. This would provide more evidence in exploring trastuzumab resistance mechanisms. We have provided a detailed discussion on Line 598-607.

We would like to extend our appreciation for your constructive suggestions and continue to improve our research in future experiments.

Recommendations for the authors:

Reviewer #2 (Recommendations for the authors):

(1) Line 334: it would be helpful to clarify that JIMT1 cells are trastuzumab-resistant while SKBR3 cells are trastuzumab sensitive, especially for those not familiar with breast cancer cell lines.

Thank you for your valuable recommendations. We added the description of trastuzumab sensitive SKBR3 and trastuzumab resistant JIMT1 on Line 334-335.

(2) Figure 3: the concentrations of erastin and RSL3 should be indicated.

Thank you for your valuable recommendations. In Figure 3, the concentration of erastin was 10 μ m and RSL3 was 1 μ m. We added these details in the figure legends on Line 872-873.

(3) Figure 3: lipid peroxidation does not necessarily mean ferroptosis. Cell viability data and rescuing effects of ferroptosis inhibitors should be shown.

Thank you for your valuable recommendations. As we mentioned above, we utilized CCK8 tests to compare cell viabilities of JIMT1 and SKBR3 in different erastin and RSL3 concentrations, as well as different exposure time of cysteine starvation. It was consistent with lipid peroxidation tests that JIMT1 was more sensitive to erastin and RSL3, but tolerant to cysteine starvation. This data was included in Figure S5C-E. We added the description in Line 375-379.

As described above, we also performed experiments to explore the rescuing effects of ferroptosis inhibitor Fer-1. It was indicated that Fer-1 could suppress the lipid peroxidation resulted from erastin, RSL3 and cysteine starvation in both JIMT1 and SKBR3. This provided more evidence that cysteine metabolism played a vital role in modulating HER2 positive breast cancer ferroptosis. This data was included in Figure S5G and S5H. We added the description to Line 387-391.

(4) Figure 3H: how cysteine starvation was performed should be clarified in the Methods section.

Thank you for your valuable recommendations. We performed cell culture with cysteine starvation by utilizing cystine/cysteine-deficient DMEM (BIOTREE) and 1% penicillin streptomycin at 37°C with 5% CO₂. We added details of this diet on Line 141-143 in Methods.

(5) Figure 4: the meaning of "H" should be clarified.

Thank you for your valuable recommendations. H was indicated as trastuzumab. We clarified the meaning of "H" in the figure legends on Line 898.

(6) Figure 4B & 4C: the data of "H" group and "Erastin" group are inconsistent.

Thank you for your valuable recommendations. In the vivo experiments, the tumor volume changes were analyzed using a paired approach, comparing the tumor size of each individual mouse before and after treatment. We noticed the confusion caused and added more details about our vivo experiments on Line 240 in Methods and Line 892-893 in figure legends.

(7) Figure 4: how cysteine starvation was performed should be clarified in the Methods section.

Thank you for your valuable recommendations. We performed cysteine starvation by utilizing cystine/cysteine-deficient diet (Xietong Bio). We added details of this diet on Line 236-237 in Methods.

We have also corrected some grammatical errors in the manuscript and We would like to extend our great appreciation to all editors and reviewers for their invaluable contributions.

<https://doi.org/10.7554/eLife.103953.2.sa0>